Group Theory

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1 Introduction

2 Abelianity and normality

Theorem 2.1: Subgroup of abelian groups

Every subgroup of an abelian group is normal.

There is some subtlety under the two cases: • G is an abelian group, • the subgroup H of G is abelian. For the first case, we can tell H is normal by theorem 2.1. But for the second case, there is no guarantee that G is abelian so we cannot use the theorem, hence we have no anticipation on whether H is normal or not. Theorem 2.1 gives us an illusion that if H is abelian, then H is normal. In fact, being normal as a subgroup does not coincide with being abelian as a subgroup:
Let G be a group and $H \leq G$. H is abelian $\not\Rightarrow$ normal and H is normal $\not\Rightarrow$ abelian.
For the first part, a counter example is to take $G = S_3$ and $H = \langle(12)\rangle$. For the second part, a counter example is $H = A_4 \trianglelefteq S_4 = G$. Note that A_4 is not abelian.
So, we can give a definition of abelian normal group.

3 p-groups

4 Group actions

Definition 4.1

Transitive group actions.

Example • Transitive group actions:
• In-transitive group actions:

4.1 Study of p-groups and Sylow-p subgroups

Theorem 4.1: Existence of subgroups of order p

Let p be a prime and let G be a finite subgroup. $\exists H \leq_{\mathbf{Gp}} G$ with $|H| = p \Leftrightarrow p\Big| \, |G|$.
Under this case, the number of subgroup of order p, $|\{H \leq G : |H| = p\}| \equiv 1 \pmod p$

Before proving this theorem, we notice that one direction is easy and another one is not easy. How to extract a subgroup of order? Let's do some example first. Consider the cyclic group of order 4, $C_4 = \langle g\rangle$ and the Klein group $\{e, a, b, c\}$. Both are 2-groups. In C_4, there is a subgroup of order 2, $\{1, g\}$. In the Klein group, there are 3 subgroups of order 2, $\{e, a\}, \{e, b\}, \{e, c\}$. All these subgroups are cyclic and are generated by elements of order 2. So, these two examples suggest us how to construct a subgroup of order p. More than that, it also suggests the number of subgroups of order p is relevant to the number of elements of order p. They might not be equal, because each such subgroup looks disjoint with another one. But at least these two numbers are closed related.
Proof: Suppose the existence of a subgroup $H \leq G$ with $|H| = p$, then $|G| = [G : H] \cdot |H|$ and $[G : H]$ is an integer. So, $p\Big| \, |G|$.

For another direction, if $p\Big| \, |G|$. We claim that $\Big|\{g \in G : |g| = p\}\Big| \equiv p - 1 \pmod p$. Since $p \geq 2$, this means this set is non-empty, showing the existence of such an element. Then, consider $g \in \{g \in G : |g| = p\}$ and the cyclic subgroup $\langle g\rangle$. $\langle g\rangle$ is a group of order p.

To show this, consider the group action $C_p \curvearrowright \{(x_1, x_2, \ldots, x_p) : x_1x_2 \cdots x_p = 1\}$. This set includes the $x \in G$ with $x^p = 1$ as a special case: those orbits of order 1. The equation $x_1x_2 \cdots x_p = 1$ has $|G|^{p-1}$ many solutions because $x_1, \ldots, x_{p-1}$ can be chosen arbitrarily and then pick $x_p = (x_1 \cdots x_{p-1})^{-1}$.

Since $|C_p| = p$, orbits have orders either 1 or p. Let k be the number of orbits of order 1. Let m be the number of orbits of order p. Then, by the theorem, $|G|^{p-1} = 1 \cdot k + p \cdot m$. Then, $|G|^{p-1} \equiv 0 \pmod p$ implies that $k \equiv 0 \pmod p$.

In the meanwhile, $k = \left|\{x \in G : x^p = 1\}\right|$. When $x^p = 1$, $|x|\Big|p$. Either $x = \mathrm{id}$ or $|x| = p$. Let s be the number elements of order p. Hence, $k = 1 + s$ and then $s \equiv p - 1 \pmod p$. □

Remark The reason why the action uses C_p instead of S_p is that under the action of C_p, it gives some simple orbits. Since every orbit divides the order of group, here it actually has two kinds of orbits, of order 1 and of order p. C_p makes the action more precise and simpler than S_p. If insisting on S_p, the proof could still work. But, it entails more argument and discussion on orbits of order different from 1 and p.

Proposition 4.1: Center of p-group

A non-trivial p-group has a non-trivial center.

Proof: Let G be a non-trivial p-group. A p-group is a group of order of p^n for some $n \geq 0$. Non-trivial means that $n \geq 1$. Suppose that the center $Z(G)$ is trivial, i.e. $Z(G) = \{e\}$. Let $\mathrm{cl}(z) := \{gzg^{-1} : g \in G\}$ be the set of conjugation element of z. Notice that

$$z \in Z(G) \Leftrightarrow \forall g \in G,\ gzg^{-1} = z \Leftrightarrow \mathrm{cl}(z) = \{z\} \Leftrightarrow |\mathrm{cl}(z)| = 1 \quad (*)$$

Hence, $Z(G) = \bigcup_{z \in G, |\mathrm{cl}(z)|=1} \mathrm{cl}(z)$. Consider the conjugate action of G on $\{x \in G : |x| = p\}$. In fact, $\mathrm{cl}(z) = \mathrm{Orb}_G(z)$ under this action. $|\mathrm{cl}(z)| = |\mathrm{Orb}_G(z)| = [G : C_G(z)]$ by orbit-stabilizer theorem.

(First version) Since G is a p-group so then $[G : C_G(z)] = p^k$ for some $k \geq 0$. $\{x \in G : |x| = p\}$ can be decomposed into orbits. G is a non-trivial p-group so $p\Big|\,|G|$. From the proof of theorem 4.1, $|\{x \in G : |x| = p\}| \equiv p - 1 \pmod p$. So, it is not possible that every orbit has size $\geq p$. There must exist an orbit of size 1. By $(*)$, this equals to $x \in Z(G)$ and $\mathrm{cl}(x) = \{x\} \subseteq \{x \in G : |x| = p\}$. This means that there exists a non-trivial element that lies in $Z(G)$. So, $Z(G)$ is non-trivial, contradicting to the assumption.

(Second version) due to Tony Varilly– decomposing the group G:
$G = \bigsqcup_i \mathrm{Orb}_G(z_i) = Z(G) \sqcup \bigsqcup_{i, |\mathrm{cl}(z_i)| \geq p} \mathrm{cl}(z_i)$. So,

$$p^n = |G| = |Z(G)| + \sum_{i, |\mathrm{cl}(z_i)| \geq p} |\mathrm{cl}(z_i)| = |Z(G)| + p \cdot k = 1 + pk$$

for some k, which leads to a contradiction. □

Theorem 4.2: Information on Sylow p-subgroup

Let G be a finite group with $|G| = p^k q$ where p is a prime and $p \not| q$. Then,
(1) **Existence**: $\mathrm{Syl}_p(G)$ exists.
(2) **Number**: The number of Sylow p-subgroup of G is n, satisfying $n \equiv 1 \pmod p$ and $n\Big||G|$
(3) **Relation**: Any two Sylow p-groups are conjugate.

Proof: □

4.2 Interaction of simple groups with p-groups

Theorem 4.3

If G is a simple group of order less than 60, then $|G|$ is prime.

Proposition 4.2

Let p be a prime and let G be a finite group with $p\Big|\,|G|$. Then,
(1) G has at least one Sylow p-group Redundant
(2) If P is the only Sylow p-group, then $P \trianglelefteq G$.

Proof: Let P be the unique Sylow p-subgroup of G. Then, $\forall g \in G$, consider gPg^{-1}. Since $gPg^{-1} \leq G$ and has the same order as P. It is another Sylow p-subgroup of G. By uniqueness, $gPg^{-1} = P$ for all g. So, P is automatically normal in G. □

Example ([McK13]) Let G be a group of order 15 and consider its Sylow 5-subgroup. Suppose there are n Sylow 5-subgroups. By 4.2, $n = 1, 6, 11, \dots$ and $n|15$. So, the only possibility is $n = 1$. So, there exists the unique Sylow 5-subgroup of G and by this theorem it is normal. Thus, G has a normal subgroup od order 5.

Proposition 4.3

Let p, q be primes with $p < q$. If G is a finite group with $|G| = pq$, then G has a normal subgroup of order q. In particular, G is not simple.

Proof: Let n_q be the number of Sylow q-groups of G. Then, by Theorem 4.2, $n_q \equiv 1 \pmod{p}$ and $n_q | pq$. Then, $n_q \nmid q$. This implies $n_q | p$. If $n_q > 1$, then $n_q > q > p$, contradicting to $n_q | p$. So, the only possibility is $n_q = 1$. By Proposition 4.3, the only Sylow p-subgroup of G is normal in G. $\square$

5 Solvable groups

Definition 5.1: Solvable, Supersolvable and Nilpotent groups

Let G be a group.
G is **solvable** $\Leftrightarrow \exists$ a sequence

$$\{1\} = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_n = G$$

(which means $\forall i$, $1 \le i \le n$, $G_{i-1} \trianglelefteq G_i$) and $\forall i$, G_i/G_{i-1} is an <u>abelian</u> group.
G is **supersolvable** $\Leftrightarrow \exists$ a sequence

$$\{1\} = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_n = G$$

(which means $\forall i$, $1 \le i \le n$, $G_{i-1} \trianglelefteq G_i$) and $\forall i$, G_i/G_{i-1} is a <u>cyclic</u> group.
G is **nilpotent** $\Leftrightarrow \exists$ a central series, i.e. a sequence of normal subgroups

$$\{1\} = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_n = G$$

(which means $\forall i$, $1 \le i \le n$, $G_{i-1} \trianglelefteq G_i$) and $\forall i$, $G_i/G_{i-1} \subseteq Z(G/G_{i-1})$. i.e. every G_i/G_{i-1} lies in the center of G/G_{i-1}.

Remark For a nilpotent group G, the integer $\min\{n | G$ has a central series of length $n\}$ is called the **nilpotency class of** G, and G is said to be **nilpotent of class** n. The length is one less than the number of groups in a chain, i.e. if the chain is $\{\mathrm{id}\} = G_0 \subseteq \cdots \subseteq G_n = G$, the length is n, but there are $n+1$ groups.

The following theorems, 5.1 and 5.3, give a chain of relations:

$$\boxed{p-\text{group} \overset{5.3}{\subseteq} \text{nilpotent group} \overset{5.1}{\subseteq} \text{supersolvable group} \subseteq \text{solvable group}}$$

The rightmost inclusion is trivial. The two left are not that immediate.
Actually we can fill more information on this chain:

$$\text{cyclic group} \subseteq \text{abelian group} \overset{5.2}{\subseteq} \text{nilpotent group} \supseteq p-\text{group}$$

For abelian groups and p-groups, they do not belong to each other. i.e. $\exists$ abelian groups that are not p-groups, for instance, cyclic group of order 6. Also, $\exists$ p-groups that are not abelian, for example, the dihedral group D_4 and quaterion group Q. Both are of order $8 = 2^3$. These examples can be made longer by adding more parts in the product, say $\mathbb{Z}/2\mathbb{Z} \times D_4$ and $\mathbb{Z}/2\mathbb{Z} \times Q$, and so on.

Theorem 5.1: Supersolvable and nilpotent

Every nilpotent group is supersolvable.

Proof: $\square$

Theorem 5.2: Nilpotency of abelian groups

Every abelian group is nilpotent.

Proof: Let G be an arbitrary abelian group. The central series always exists because $\{1\} \trianglelefteq G$ exists. So, for every central series $\{1\} = G_0 \subseteq G_1 \subseteq \cdots \subseteq G_n = G$, $\forall i$, $G_{i+1}/G_i \subseteq G/G_i = Z(G/G_i)$ since G is abelian. $\square$

Theorem 5.3: p group and nilpotent

Every p-group of finite order is nilpotent.

Proof: Pick an arbitrary p-group G. Induct on $|G|$:
Base case: $|G| = 1$, then $G = \{e\}$ and G is nilpotent.
Inductive hypothesis: $|G| =: n > 1$, $\forall$ p-group of order $< n$ are nilpotent.
Inductive step: Since G is a non-trivial p-group by assumption, then $Z(G)$ is nontrivial by Lemma 5.2.
From theorem 5.2, $Z(G)$ is abelian. Hence it is nilpotent.
Since $Z(G)$ is non-trivial, then $|G/Z(G)| < |G|$. The quotient group $G/Z(G)$ is a p-group of order $|G/Z(G)| < |G|$.
By inductive hypothesis, $G/Z(G)$ is nilpotent.
By Lemma 5.3, since $G/Z(G)$ and $Z(G)$ are nilpotent, G is nilpotent. $\square$
To prove 5.2 and other things, we need the following lemma [Ser77]

Lemma 5.1: Property of fixed set of a p-group

Let G be a p-group and G acting on a finite set X. X^G is the set of elements of X fixed by G. Then,

$$|X| \equiv |X^G| \mod p$$

Lemma 5.2: Non-trivial center for p-group

Let G be a nontrivial p-group. Then $Z(G)$ is nontrivial.

Lemma 5.3: Property for nilpotent group

Let G be a group. If $N \leq Z(G)$ and G/N is nilpotent, then G is nilpotent.

References

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